



Submitted to:-

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Section:-

5 A

Cloud Computing
Project

Project 1 – Multi-Environment Static Website (S3 + CloudFront) with Terraform & Ansible

<https://github.com/Hafsa-SE/Project1>

Required installations

```
● @Hafsa-SE → /workspaces/Project1 (main) $ aws --version
aws-cli/2.33.6 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
○ @Hafsa-SE → /workspaces/Project1 (main) $
```

```
● @Hafsa-SE → /workspaces/Project1 (main) $ git --version
git version 2.52.0
○ @Hafsa-SE → /workspaces/Project1 (main) $
```

```
● @Hafsa-SE → /workspaces/Project1 (main) $ terraform -version
Terraform v1.14.3
on linux_amd64
○ @Hafsa-SE → /workspaces/Project1 (main) $
```

```
● @Hafsa-SE → /workspaces/Project1 (main) $ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Nov 6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
○ @Hafsa-SE → /workspaces/Project1 (main) $
```

```
● @Hafsa-SE → /workspaces/Project1 (main) $ python3 --version
Python 3.12.1
● @Hafsa-SE → /workspaces/Project1 (main) $ pip3 --version
pip 25.3 from /usr/local/python/3.12.1/lib/python3.12/site-packages/pip (python 3.12)
○ @Hafsa-SE → /workspaces/Project1 (main) $
```

```
● @Hafsa-SE → /workspaces/Project1 (main) $ ansible-galaxy collection list | grep amazon.aws
amazon.aws 7.2.0
○ @Hafsa-SE → /workspaces/Project1 (main) $
```

Part 1: Terraform Infrastructure Setup

1.1 Project Structure

```
● @Hafsa-SE → /workspaces/Project1 (main) $ mkdir Project1
  cd Project1
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ touch main.tf variables.tf outputs.tf locals.tf
  terraform.tfvars README.md .gitignore
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ mkdir -p modules/s3_site modules/cloudfront
  touch modules/s3_site/main.tf modules/s3_site/variables.tf modules/s3_site/outputs.tf
  touch modules/cloudfront/main.tf modules/cloudfront/variables.tf modules/cloudfront/outputs.tf
○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $
```

Create Ansible Structure

```
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ mkdir -p ansible/inventory ansible/playbooks
  touch ansible/ansible.cfg ansible/inventory/localhost.yml ansible/playbooks/sync-site.yml

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ mkdir -p site/assets
  touch site/index.html
○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $
```

Implement proper .gitignore

```
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ nano .gitignore
```

```
GNU nano 7.2 .gitignore *
# Terraform
.terraform/
*.tfstate
*.tfstate.backup
terraform.tfvars

# OS files
.DS_Store
Thumbs.db

# Editor files
.vscode/
.idea/

# Logs
*.log
```

1.2 Variable Configuration

```
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ nano variables.tf
```

```

GNU nano 7.2                                     variables.tf *
variable "aws_region" {
  description = "AWS region where resources will be created"
  type        = string
}

variable "env" {
  description = "Deployment environment (dev, staging, prod)"
  type        = string

  validation {
    condition     = contains(["dev", "staging", "prod"], var.env)
    error_message = "env must be one of: dev, staging, prod."
  }
}

variable "project_name" {
  description = "Project name used for resource naming and tagging"
  type        = string
}

variable "bucket_prefix" {
  description = "Prefix for S3 bucket name"
  type        = string
}

variable "enable_logging" {
  description = "Enable CloudFront access logging"
  type        = bool
  default     = false
}

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano terraform.tfvars

```

GNU nano 7.2                                     terraform.tfvars *
aws_region   = "me-central-1"
env          = "dev"
project_name = "cc-static-site"
bucket_prefix = "cc-static"
enable_logging = false

tags = {
  Owner = "Hafsa"
  Class = "CloudComputing"
}

```

1.3 Locals Configuration

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano locals.tf

```
GNU nano 7.2                                locals.tf *
locals {
  common_tags = merge(
    {
      Project      = var.project_name
      Environment  = var.env
      ManagedBy    = "Terraform"
    },
    var.tags
  )

  bucket_name = lower("${var.bucket_prefix}-${var.project_name}-${var.env}")

  cloudfront_comment = "Static site for ${var.project_name} (${var.env})"
}
```

1.4 S3 Site Module

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano modules/s3_site/variables.tf

```
GNU nano 7.2                                modules/s3_site/variables.tf *
variable "bucket_name" {
  description = "Name of the S3 bucket"
  type        = string
}

variable "tags" {
  description = "Tags applied to S3 bucket"
  type        = map(string)
}
```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano modules/s3_site/main.tf

```

GNU nano 7.2                                     modules/s3_site/main.tf *
resource "aws_s3_bucket" "this" {
  bucket = var.bucket_name
  tags   = var.tags
}

resource "aws_s3_bucket_public_access_block" "this" {
  bucket = aws_s3_bucket.this.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls     = true
  restrict_public_buckets = true
}

resource "aws_s3_bucket_policy" "this" {
  bucket = aws_s3_bucket.this.id

  policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Effect    = "Deny"
        Principal = "*"
        Action     = "s3:GetObject"
        Resource   = "${aws_s3_bucket.this.arn}/*"
      }
    ]
  })
}

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano modules/s3_site/outputs.tf

```

GNU nano 7.2                                     modules/s3_site/outputs.tf *
output "bucket_name" {
  value = aws_s3_bucket.this.bucket
}

output "bucket_arn" {
  value = aws_s3_bucket.this.arn
}

output "bucket_domain_name" {
  value = aws_s3_bucket.this.bucket_regional_domain_name
}

```

Part 1 Testing

- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ terraform init -backend=false
Initializing provider plugins...

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ terraform validate
Success! The configuration is valid.

Part 2: CloudFront Module

2.1 Module Design

- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ mkdir -p modules/cloudfront
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ ls modules
cloudfront s3_site
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ touch modules/cloudfront/variables.tf
touch modules/cloudfront/main.tf
touch modules/cloudfront/outputs.tf
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano modules/cloudfront/variables.tf

GNU nano 7.2

modules/cloudfront/variables.tf *

```
variable "project_name" {  
  description = "Name of the project"  
  type        = string  
}  
  
variable "env" {  
  description = "Environment name (dev/staging/prod)"  
  type        = string  
}  
  
variable "bucket_domain_name" {  
  description = "S3 bucket domain name for CloudFront origin"  
  type        = string  
}  
  
variable "comment" {  
  description = "CloudFront distribution comment"  
  type        = string  
}  
  
variable "enable_logging" {  
  description = "Enable CloudFront logging"  
  type        = bool  
  default     = false  
}  
  
variable "logging_bucket" {  
  description = "S3 bucket for CloudFront logs"  
  type        = string  
  default     = ""  
}
```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano modules/cloudfront/main.tf


```

GNU nano 7.2                                modules/cloudfront/main.tf *
resource "aws_cloudfront_distribution" "this" {

    enabled          = true
    comment          = var.comment
    default_root_object = "index.html"

    origin {
        domain_name = var.bucket_domain_name
        origin_id    = "${var.project_name}-${var.env}-s3-origin"
    }

    default_cache_behavior {
        allowed_methods      = ["GET", "HEAD"]
        cached_methods       = ["GET", "HEAD"]
        target_origin_id     = "${var.project_name}-${var.env}-s3-origin"
        viewer_protocol_policy = "redirect-to-https"

        forwarded_values {
            query_string = false
            cookies {
                forward = "none"
            }
        }
    }

    restrictions {
        geo_restriction {
            restriction_type = "none"
        }
    }
}

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano modules/cloudfront/outputs.tf

```

GNU nano 7.2                                modules/cloudfront/outputs.tf *
output "distribution_id" {
    description = "CloudFront distribution ID"
    value       = aws_cloudfront_distribution.this.id
}

output "domain_name" {
    description = "CloudFront distribution domain name"
    value       = aws_cloudfront_distribution.this.domain_name
}

```

2.2 Module Usage

● @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano main.tf

```

GNU nano 7.2                                main.tf *
module "cdn" {
  source           = "./modules/cloudfront"
  project_name     = var.project_name
  env              = var.env
  bucket_domain_name = module.s3_site.bucket_domain_name
  comment          = "Static site for ${var.project_name}-${var.env}"
  enable_logging    = var.enable_logging
  logging_bucket    = var.logging_bucket
  tags              = local.common_tags
}

```

Part 2 Testing

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ terraform init
  Initializing the backend...
  Initializing modules...
  - cdn in modules/cloudfront
  Initializing provider plugins...
  - Finding latest version of hashicorp/aws...
  - Installing hashicorp/aws v6.28.0...
  - Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```

○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $ █

```

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ terraform validate
  Success! The configuration is valid.

```

```

○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $ █

```

```
@Hafsa-SE → /workspaces/Project1/Project1 (main) $ terraform plan
```

```
+ server_side_encryption_configuration (known after apply)
+ versioning (known after apply)
+ website (known after apply)
}

# module.s3_site.aws_s3_bucket_policy.this will be created
+ resource "aws_s3_bucket_policy" "this" {
  + bucket = (known after apply)
  + id      = (known after apply)
  + policy = (known after apply)
  + region = "me-central-1"
}

# module.s3_site.aws_s3_bucket_public_access_block.this will be created
+ resource "aws_s3_bucket_public_access_block" "this" {
  + block_public_acls      = true
  + block_public_policy    = true
  + bucket                 = (known after apply)
  + id                     = (known after apply)
  + ignore_public_acls     = true
  + region                 = "me-central-1"
  + restrict_public_buckets = true
}
```

Plan: 4 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $
```

Part 3: Ansible Content Deployment

3.1 Ansible Setup

```
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ mkdir -p ansible/inventory
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ nano ansible/ansible.cfg
○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $
```

```
GNU nano 7.2                                ansible/ansible.cfg *
[defaults]
inventory = inventory/localhost.yml
host_key_checking = False
retry_files_enabled = False
stdout_callback = yaml
```

```
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ nano ansible/inventory/localhost.yml
```

```
GNU nano 7.2                                ansible/inventory/localhost.yml *
```

```
all:
  hosts:
    localhost:
      ansible_connection: local
```

```
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ cd ansible
● @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) $ ansible-inventory -i inventory/localhost.yml --list
[WARNING]: Ansible is being run in a world writable directory
(/workspaces/Project1/Project1/ansible), ignoring it as an ansible.cfg source. For more
information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
{
  "_meta": {
    "hostvars": {
      "localhost": {
        "ansible_connection": "local"
      }
    }
  },
  "all": {
    "children": [
      "ungrouped"
    ]
  },
  "ungrouped": {
    "hosts": [
      "localhost"
    ]
  }
}
```

```
○ @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) $
```

3.2 sync-site.yml Playbook

```
● @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) $ mkdir -p playbooks
● @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) $ nano playbooks/sync-site.yml
```

```

GNU nano 7.2                                playbooks/sync-site.yml *
--
- name: Sync static site to S3
  hosts: localhost
  gather_facts: false

  vars:
    bucket_name: "{{ bucket_name | default(lookup('env', 'BUCKET_NAME')) }}"
    site_dir: "../../site"

  tasks:
    - name: Ensure bucket name provided
      assert:
        that:
          - bucket_name is defined
          - bucket_name | length > 0
        fail_msg: "bucket_name is required (pass via --extra-vars or BUCKET_NAME env)."

    - name: Sync site content to S3
      amazon.aws.s3_sync:
        region: "{{ lookup('env', 'AWS_REGION') | default('us-east-1') }}"
        bucket: "{{ bucket_name }}"
        file_root: "{{ site_dir }}"
        delete: false

```

Use `bucket_name` and `AWS_REGION` from environment or extra-vars

```

● @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) $ export AWS_REGION=me-central-1
● @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) $ export BUCKET_NAME=cc-static-cc-static-site-dev
○ @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) $ █

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ ANSIBLE_CONFIG=ansible/ansible.cfg ansible-playbook ansible/playbooks/sync-site.yml

PLAY [Sync static site to S3] *****

TASK [Ensure bucket name provided] *****
ok: [localhost] => changed=false
   msg: All assertions passed

TASK [Sync site content using AWS CLI] *****
changed: [localhost]

PLAY RECAP *****
localhost                : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescue
d=0    ignored=0

○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $ █

```

Part 3 Testing

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ aws s3 ls s3://$BUCKET_NAME --recursive
2026-01-26 11:37:19          0 index.html
○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $ █

```

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ cat > site/index.html << 'EOF'
<!DOCTYPE html>
<html>
<head>
  <title>My Static Site</title>
</head>
<body>
  <h1>Welcome to Hafsa's Static Website</h1>
  <p>This is a test page for Cloud Computing project.</p>
</body>
</html>
EOF

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ ANSIBLE_CONFIG=ansible/ansible.cfg ansible-play
book ansible/playbooks/sync-site.yml --extra-vars "bucket_name=$BUCKET_NAME"

PLAY [Sync static site to S3] *****

TASK [Ensure bucket name provided] *****
ok: [localhost] => changed=false
   msg: All assertions passed

TASK [Sync site content using AWS CLI] *****
changed: [localhost]

PLAY RECAP *****
localhost          : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescue
d=0    ignored=0

○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $ █

```

Part 4: Deployment & Testing

4.1 Terraform Deployment

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ terraform init
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0

```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```

○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $ █

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ terraform validate
Success! The configuration is valid.

```

```
@Hafsa-SE →/workspaces/Project1/Project1 (main) $ terraform plan -var="env=dev"
```

```
+ server_side_encryption_configuration (known after apply)
+ versioning (known after apply)
+ website (known after apply)
}

# module.s3_site.aws_s3_bucket_policy.this will be created
+ resource "aws_s3_bucket_policy" "this" {
  + bucket = (known after apply)
  + id     = (known after apply)
  + policy = (known after apply)
  + region = "me-central-1"
}

# module.s3_site.aws_s3_bucket_public_access_block.this will be created
+ resource "aws_s3_bucket_public_access_block" "this" {
  + block_public_acls       = true
  + block_public_policy     = true
  + bucket                 = (known after apply)
  + id                     = (known after apply)
  + ignore_public_acls     = true
  + region                 = "me-central-1"
  + restrict_public_buckets = true
}
```

Plan: 4 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
○ @Hafsa-SE →/workspaces/Project1/Project1 (main) $
```

```
● @Hafsa-SE →/workspaces/Project1/Project1 (main) $ terraform apply -auto-approve -var="env=dev"
module.s3_site.aws_s3_bucket.this: Refreshing state... [id=cc-static-cc-static-site-dev]
module.s3_site.aws_s3_bucket_public_access_block.this: Refreshing state... [id=cc-static-cc-static-site-dev]
module.s3_site.aws_s3_bucket_policy.this: Refreshing state... [id=cc-static-cc-static-site-dev]
module.cdn.aws_cloudfront_distribution.this: Refreshing state... [id=E1Z5BTBxB0JYGT]
```

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

```
○ @Hafsa-SE →/workspaces/Project1/Project1 (main) $
```

4.2 Terraform Outputs

```
● @Hafsa-SE →/workspaces/Project1/Project1 (main) $ nano outputs.tf
```

```

GNU nano 7.2                                outputs.tf *
output "bucket_name" {
    description = "S3 bucket name for static site"
    value       = module.s3_site.bucket_name
}

output "cloudfront_domain_name" {
    description = "CloudFront distribution domain name"
    value       = module.cdn.domain_name
}

```

- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ terraform output
 bucket_name = "cc-static-cc-static-site-dev"
 cloudfront_domain_name = "d3sd7t52jjiwxb.cloudfront.net"
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ export BUCKET_NAME=\$(terraform output -raw bucket_name)
 export AWS_REGION=me-central-1
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ echo \$BUCKET_NAME
 cc-static-cc-static-site-dev
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ echo \$AWS_REGION
 me-central-1
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$

4.3 Ansible Deployment

- @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) \$ ansible-playbook playbooks/sync-site.yml --extra-vars "bucket_name=\$BUCKET_NAME"
- ```

PLAY [Sync static site to S3] *****

TASK [Ensure bucket name provided] *****
ok: [localhost] => changed=false
 msg: All assertions passed

TASK [Sync site content using AWS CLI] *****
changed: [localhost]

PLAY RECAP *****
localhost : ok=2 changed=1 unreachable=0 failed=0 skipped=0
rescued=0 ignored=0

```
- @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) \$



aws Search [Alt+S] Middle East (UAE) Hafsa Khalid (7647-6688-9734) Hafsa%20Khalid

Amazon S3

**Amazon S3**

- ▼ Buckets
  - General purpose buckets
  - Directory buckets
  - Table buckets
  - Vector buckets
- ▼ Access management and security
  - Access Points

**General purpose buckets** All AWS Regions Directory buckets

**General purpose buckets (1)** Info

Buckets are containers for data stored in S3.

Find buckets by name

| Name                         | AWS Region                     | Creation date                          |
|------------------------------|--------------------------------|----------------------------------------|
| cc-static-cc-static-site-dev | Middle East (UAE) me-central-1 | January 26, 2026, 16:51:20 (UTC+05:00) |

aws Search [Alt+S] Middle East (UAE) Hafsa Khalid (7647-6688-9734) Hafsa%20Khalid

Amazon S3 > Buckets > cc-static-cc-static-site-dev

**Amazon S3**

- ▼ Buckets
  - General purpose buckets
  - Directory buckets
  - Table buckets
  - Vector buckets
- ▼ Access management and security
  - Access Points
  - Access Points for FSx
  - Access Grants
  - IAM Access Analyzer
- ▼ Storage management and insights
  - Storage Lens

**cc-static-cc-static-site-dev** Info

Objects Metadata Properties Permissions Metrics Management Access Points

**Objects (1)**

Copy S3 URI Copy URL Download Open Delete Actions Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

| Name       | Type | Last modified                          | Size    | Storage class |
|------------|------|----------------------------------------|---------|---------------|
| index.html | html | January 26, 2026, 17:46:02 (UTC+05:00) | 196.0 B | Standard      |

- @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) \$ aws s3 ls s3://\$BUCKET\_NAME --recursive  
2026-01-26 12:46:02 196 index.html
- @Hafsa-SE → /workspaces/Project1/Project1/ansible (main) \$

## 4.4 Website Testing

```
@Hafsa-SE → /workspaces/Project1/Project1 (main) $ terraform apply
```

```
module.cdn.aws_cloudfront_origin_access_identity.oai: Creating...
module.oai.aws_cloudfront_origin_access_identity.oai: Creating...
module.oai.aws_cloudfront_origin_access_identity.oai: Creation complete after 2s [id=E2F5KTMC5AWB3W]
module.cdn.aws_cloudfront_origin_access_identity.oai: Creation complete after 2s [id=E1HT23YNZ3FU4M]
module.s3_site.aws_s3_bucket_policy.this: Modifying... [id=cc-static-cc-static-site-dev]
module.cdn.aws_cloudfront_distribution.this: Modifying... [id=E1Z5BTBxB0JYGT]
module.s3_site.aws_s3_bucket_policy.this: Modifications complete after 9s [id=cc-static-cc-static-site-dev]
module.cdn.aws_cloudfront_distribution.this: Still modifying... [id=E1Z5BTBxB0JYGT, 00m10s elapsed]
module.cdn.aws_cloudfront_distribution.this: Still modifying... [id=E1Z5BTBxB0JYGT, 00m20s elapsed]
module.cdn.aws_cloudfront_distribution.this: Still modifying... [id=E1Z5BTBxB0JYGT, 00m30s elapsed]
module.cdn.aws_cloudfront_distribution.this: Still modifying... [id=E1Z5BTBxB0JYGT, 00m40s elapsed]
module.cdn.aws_cloudfront_distribution.this: Still modifying... [id=E1Z5BTBxB0JYGT, 00m50s elapsed]
module.cdn.aws_cloudfront_distribution.this: Still modifying... [id=E1Z5BTBxB0JYGT, 01m00s elapsed]
module.cdn.aws_cloudfront_distribution.this: Still modifying... [id=E1Z5BTBxB0JYGT, 01m10s elapsed]
module.cdn.aws_cloudfront_distribution.this: Modifications complete after 1m10s [id=E1Z5BTBxB0JYGT]
```

Apply complete! Resources: 2 added, 2 changed, 0 destroyed.

### Outputs:

```
bucket_name = "cc-static-cc-static-site-dev"
cloudfront_domain_name = "d3sd7t52jjiwxb.cloudfront.net"
```

```
@Hafsa-SE → /workspaces/Project1/Project1 (main) $
```

```
● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ terraform output
 bucket_name = "cc-static-cc-static-site-dev"
 cloudfront_domain_name = "d3sd7t52jjiwxb.cloudfront.net"
```

← → ↻ 🔍 d3sd7t52jjiwxb.cloudfront.net

# Welcome to Hafsa's Static Website

This is a test page for Cloud Computing project.

## Part 5: Security & Cleanup

### 5.1 Bucket Access Verification

- Try to access an object via direct S3 URL (should return 403)

← → ↻ ⚠ Not secure cc-static-cc-static-site-dev.s3-website.me-central-1.amazonaws.com

#### 404 Not Found

- Code: NoSuchWebsiteConfiguration
- Message: The specified bucket does not have a website configuration
- BucketName: cc-static-cc-static-site-dev
- RequestId: AHNNDTP1VB88522J
- HostId: 6dz+q4U0eWWq/pSXXXah34J14p09//hrTLbH/pdOKKaALAFqH+ETWdPhwhb/s7St9VXrsO0uFvRvVp5tkXTHC6dKitM6D+ZWa

← → ↻ 🔍 cc-static-cc-static-site-dev.s3.me-central-1.amazonaws.com/index.html

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" ?>
<Error>
 <Code>AccessDenied</Code>
 <Message>Access Denied</Message>
 <RequestId>FKEQPTN6EJZEHF4B</RequestId>
 <HostId>oFmxXl8n4rFxf0xZao2lrv0Cr+KECsPc8jNeom9puUS0UI/VJr1fr02yvtvKyMR4PjV9Sj+T5IYw1DYzpDNk8gbL2H7VOXk</HostId>
</Error>
```

- Access the same object via CloudFront URL (should return 200)

← → ↻ 🔍 d3sd7t52jjiwxb.cloudfront.net

# Welcome to Hafsa's Static Website

This is a test page for Cloud Computing project.

## 5.2 README Documentation

- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ touch README.md
- @Hafsa-SE → /workspaces/Project1/Project1 (main) \$ nano README.md

```
GNU nano 7.2 README.md
Multi-Environment Static Website using AWS, Terraform & Ansible

Project Overview
This project deploys a secure multi-environment static website on AWS using Terraform and
Three environments are supported: dev, staging, and production.

Each environment includes:
- A private S3 bucket to store static website files
- A CloudFront distribution in front of the S3 bucket
- Secure access using CloudFront Origin Access Identity (OAI)
- HTTP to HTTPS redirection via CloudFront

Direct public access to the S3 bucket is blocked. Website content is accessible only throu

Architecture Diagram

User → CloudFront (HTTPS) → Private S3 Bucket (OAI access only)

Terraform Setup & Usage

Terraform is used to create AWS infrastructure.

Initialize Terraform
terraform init

Validate configuration
terraform validate
```

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ cat README.md
Multi-Environment Static Website using AWS, Terraform & Ansible

Project Overview
This project deploys a secure multi-environment static website on AWS using Terraform and Ansible.
Three environments are supported: dev, staging, and production.

Each environment includes:
- A private S3 bucket to store static website files
- A CloudFront distribution in front of the S3 bucket
- Secure access using CloudFront Origin Access Identity (OAI)
- HTTP to HTTPS redirection via CloudFront

Direct public access to the S3 bucket is blocked. Website content is accessible only through CloudFront.

Architecture Diagram

User → CloudFront (HTTPS) → Private S3 Bucket (OAI access only)

Terraform Setup & Usage

Terraform is used to create AWS infrastructure.

Initialize Terraform
terraform init

Validate configuration
terraform validate

```

## 5.3 Cleanup

```

● @Hafsa-SE → /workspaces/Project1/Project1 (main) $ aws s3 rm s3://cc-static-cc-static-site-dev --recursive
delete: s3://cc-static-cc-static-site-dev/index.html
○ @Hafsa-SE → /workspaces/Project1/Project1 (main) $ █

```

```
@Hafsa-SE →/workspaces/Project1/Project1 (main) $ terraform destroy -var="env=dev"
```

```
}

- server_side_encryption_configuration {
 - rule {
 - bucket_key_enabled = false -> null

 - apply_server_side_encryption_by_default {
 - sse_algorithm = "AES256" -> null
 # (1 unchanged attribute hidden)
 }
 }
}

- versioning {
 - enabled = false -> null
 - mfa_delete = false -> null
}
}
```

**Plan:** 0 to add, 0 to change, 1 to destroy.

Changes to Outputs:

```
- bucket_name = "cc-static-cc-static-site-dev" -> null
```

**Do you really want to destroy all resources?**

Terraform will destroy all your managed infrastructure, as shown above.  
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

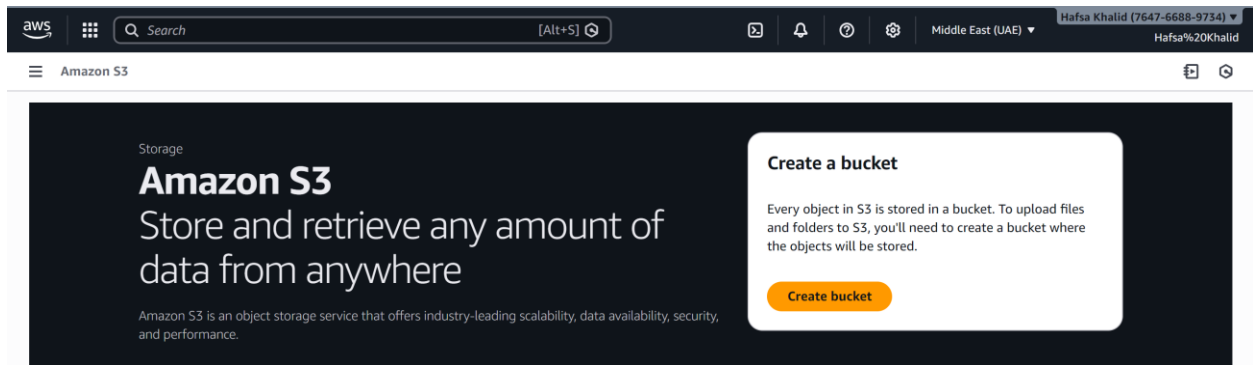
```
module.s3_site.aws_s3_bucket.this: Destroying... [id=cc-static-cc-static-site-dev]
module.s3_site.aws_s3_bucket.this: Destruction complete after 0s
```

**Destroy complete! Resources: 1 destroyed.**

```
@Hafsa-SE →/workspaces/Project1/Project1 (main) $
```

- **Confirm deletion in AWS Console**





## Repository Structure:

### ▼ PROJECT1 [CODESPACES: SHINY SPACE ZEBRA]

- > aws
- ▼ Project1
  - > .terraform
  - > ansible
  - ▼ modules
    - > cloudfront
    - > oai
    - > s3\_site
  - > site
- ◆ .gitignore
- ≡ .terraform.lock.hcl
- Y locals.tf
- Y main.tf
- Y outputs.tf
- i README.md
- ≡ terraform.tfstate
- ≡ terraform.tfstate.backup
- Y terraform.tfvars
- Y variables.tf
- awscliv2.zip
- i README.md